

# SUPPLEMENTARY MATERIALS: KEEP THE ANGLE: A GEOMETRY-PRESERVING BASIS IN SPECTRAL EMBEDDINGS

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This supplement collects the longer proofs supporting the main article: the conditional graph-to-continuum transfer theorem (Section SM1) and the truncation-uniformity results (Section SM2).<sup>1</sup> It is unrefereed supplementary material; the main article is self-contained at the level of statements, with each proof here cross-referenced from the theorem it establishes.

## SM1. Proof of Theorem 3.10.

**Setting and notation.**  $M$  is a compact  $d$ -dimensional Riemannian manifold without boundary, volume normalized to 1;  $(\mu_k, \varphi_k)_{k \geq 0}$  are the eigenpairs of its (positive) Laplace–Beltrami operator,  $0 = \mu_0 < \mu_1 \leq \mu_2 \leq \dots$ , with  $\{\varphi_k\}$  orthonormal in  $L^2(M)$ . Under non-uniform sampling with an admissible density the identical argument runs against the density-weighted operator that the graph Laplacian actually converges to [SM8, SM5] (the  $\alpha=1$  normalization of Paper I.b restores the geometric operator), with (H1)–(H2) read for its eigenpairs. The continuum objects are

$$F(x) = \left( \frac{\varphi_k(x)}{\sqrt{\mu_k}} \right)_{k=1}^m, \quad R(x) = \|F(x)\|, \quad N(x) = F(x)/R(x),$$

and  $m$  sits at a spectral gap:  $\mu_m < \mu_{m+1}$ . Write  $\Phi_m := \max_{k \leq m} \|\varphi_k\|_\infty < \infty$  (eigenfunctions on a smooth compact manifold are smooth). The sample is  $x_1, \dots, x_n$  i.i.d. from the stated density class,  $G_n$  the  $\varepsilon$ -graph (or  $k$ -NN graph) on it, and  $(\lambda_k, v_k)$  the eigenpairs of  $L_{\text{sym}}$  with the empirical normalization  $\frac{1}{n} \sum_i v_k(i)^2 = 1$ ; the algorithm forms  $\Psi_i = (v_k(i)/\sqrt{\lambda_k})_{k=1}^m$  and  $\hat{u}_i = \Psi_i / \|\Psi_i\|$ . Alongside these, the *random-walk* eigenvectors  $v_k^{\text{rw}} := D^{-1/2} v_k$  are the eigenpairs of  $L_{\text{rw}} = D^{-1}(D - A) = D^{-1/2} L_{\text{sym}} D^{1/2}$  at the *same* eigenvalue  $\lambda_k$ ; we take them with the coordinated normalization  $\frac{1}{n} \sum_i k_i v_k^{\text{rw}}(i)^2 = 1$  (the empirical degree-weighted inner product—which is *identically* the algorithm’s own  $L_{\text{sym}}$  constraint  $\frac{1}{n} \sum_i v_k(i)^2 = 1$  viewed through  $D^{1/2}$ ), under which each  $v_k^{\text{rw}}$  pairs mode-for-mode with the continuum eigenfunction  $\varphi_k$  orthonormal in the effective sampling measure (exactly  $L^2(\rho)$  for uniform sampling or after the  $\alpha=1$  reweighting of Paper I.b that restores the geometric operator; the  $\rho$ -versus- $\rho^2$  distinction under raw non-uniform sampling is immaterial,  $v_k^{\text{rw}}$  being only an intermediary to the identical angular map)—the convention in which the  $\infty$ -norm eigenvector rates below hold. Write  $\Psi_i^{\text{rw}} = (v_k^{\text{rw}}(i)/\sqrt{\lambda_k})_{k=1}^m$ .

**Step 0: three invariances.** The claim concerns only the pairwise distances  $\|\hat{u}_i - \hat{u}_j\|$ . These are invariant under (i) a global rescaling of all eigenvectors (the choice of normalization), (ii) a global rescaling of all eigenvalues (which rescales every weight  $1/\sqrt{\lambda_k}$  by the same factor), and (iii) a fixed orthogonal transformation  $Q \in O(m)$  applied to every  $\Psi_i$  (then  $\|Q\hat{u}_i - Q\hat{u}_j\| = \|\hat{u}_i - \hat{u}_j\|$ ). We use (i) to work in the empirical normalization, (ii) to apply the deterministic graph-to-continuum eigenvalue scaling  $c_n$  of [SM5], and (iii) to absorb the basis alignment of Lemma SM1.3.

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<sup>1</sup>Theorem, corollary, lemma, equation, and section numbers carrying an “SM” prefix refer to this supplement; all other cross-references point to the main article.

(iv) *Exact per-node degree cancellation.* Since  $v_k = D^{1/2}v_k^{\text{rw}}$  applies the *same* node factor  $\sqrt{k_i}$  in every mode  $k$ , the two commute-weighted embeddings differ only by a positive per-node scalar,

$$\Psi_i = (v_k(i)/\sqrt{\lambda_k})_{k=1}^m = \sqrt{k_i} (v_k^{\text{rw}}(i)/\sqrt{\lambda_k})_{k=1}^m = \sqrt{k_i} \Psi_i^{\text{rw}},$$

so row normalization returns the *identical* angular map  $\hat{u}_i = \Psi_i^{\text{rw}}/\|\Psi_i^{\text{rw}}\|$ —exactly, with no degree-ratio hypothesis. We may therefore calibrate the random-walk embedding  $\Psi_i^{\text{rw}}$ , whose  $\infty$ -norm convergence to the continuum is exactly the cited theorem, and read off the angle the algorithm computes from  $L_{\text{sym}}$ . This is what lets the theorem hold under non-uniform sampling: the node-dependent degree factor a symmetric calibration would have to control is cancelled by the normalization, not bounded.

**Step 1: the continuum angular map is locally bi-Lipschitz.**

LEMMA SM1.1. *Under (H1)–(H2), for all  $x, y \in M$  with  $d_M(x, y) \leq r_0$ ,*

$$a d_M(x, y) \leq \|N(x) - N(y)\| \leq b d_M(x, y), \quad a := \frac{\sqrt{L_0^{-2} - \Lambda_0^2}}{\rho^+}, \quad b := \frac{L_0}{\rho^-}.$$

*Proof.* Apply the identity of Lemma 3.6 to  $u = F(x)$ ,  $v = F(y)$ . By (H1),  $L_0^{-2}d_M^2 \leq \|F(x) - F(y)\|^2 \leq L_0^2d_M^2$ ; by (H2),  $(R(x) - R(y))^2 \leq \Lambda_0^2d_M^2$  and  $(\rho^-)^2 \leq R(x)R(y) \leq (\rho^+)^2$ . The ratio in the identity is therefore at least  $(L_0^{-2} - \Lambda_0^2)d_M^2/(\rho^+)^2$ —positive precisely because  $\Lambda_0 < 1/L_0$ —and at most  $L_0^2d_M^2/(\rho^-)^2$ .  $\square$

Note  $a \leq b$ : a two-sided constant satisfies  $L_0^{-1} \leq L_0$ , so  $L_0 \geq 1$ , hence  $\sqrt{L_0^{-2} - \Lambda_0^2} \leq L_0^{-1} \leq L_0$ , and  $\rho^- \leq \rho^+$ .

**Step 2: normalization is stable.**

LEMMA SM1.2. *For nonzero  $u, w \in \mathbb{R}^m$ ,*

$$\left\| \frac{u}{\|u\|} - \frac{w}{\|w\|} \right\| \leq \frac{2\|u - w\|}{\max(\|u\|, \|w\|)}.$$

*Proof.*  $\frac{u}{\|u\|} - \frac{w}{\|w\|} = \frac{u-w}{\|u\|} + w\left(\frac{1}{\|u\|} - \frac{1}{\|w\|}\right)$ , and the second term has norm  $\|w\| - \|u\|/\|u\| \leq \|u - w\|/\|u\|$ . So the left side is at most  $2\|u - w\|/\|u\|$ ; by symmetry in  $u \leftrightarrow w$ , also at most  $2\|u - w\|/\|w\|$ .  $\square$

**Step 3: sup-norm spectral transfer.** This is the step for which  $L^2$  consistency [SM8] does not suffice and the  $\infty$ -norm rates of [SM5] are the right tool. Group  $\mu_1 \leq \dots \leq \mu_m$  into blocks of equal eigenvalues (multiplets); because  $m$  sits at a spectral gap, the last block ends exactly at  $m$ .

Throughout this step the calibration is carried out on the *random-walk* eigenvectors: to keep notation light we drop the superscript and write  $v_k, \Psi_i, \tilde{\Psi}_i$  for  $v_k^{\text{rw}}$  and the corresponding random-walk commute embeddings. By invariance (iv) the resulting angle is exactly the algorithm's  $\hat{u}_i$ , and the  $\infty$ -norm rates of [SM5]—stated for the unnormalized/random-walk eigenvectors—apply to  $v_k$  directly, with no degree-ratio hypothesis.

LEMMA SM1.3 (Calibration; external sup-norm transfer). *Under the sampling model of [SM5, SM6] there are sequences  $\epsilon_n, \delta_n \rightarrow 0$ —the eigenvector  $L^\infty$  rate of [SM6] (which establishes high-probability  $L^\infty$  and  $C^{0,1}$  convergence of graph Laplacian eigenvectors to the Laplace–Beltrami eigenfunctions, with constants depending on the eigenvalue  $\mu_m$  and hence on  $m$ ;  $\epsilon_n$  carries a factor  $\sqrt{m}$ ), and the eigenvalue and gap-stability rate  $\delta_n$  of [SM5]—and events  $\mathcal{A}_n$  with  $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$  on which, after the eigenvalue scaling  $\tilde{\lambda}_k := \lambda_k/c_n$  of invariance (ii), there is an orthogonal  $Q_n \in O(m)$ , block-diagonal across the multiplets, with*

$$\max_{k \leq m} |\tilde{\lambda}_k - \mu_k| \leq \delta_n, \quad \max_{i \leq n} \|Q_n v(i) - \varphi(x_i)\| \leq \epsilon_n,$$

where  $v(i) = (v_1(i), \dots, v_m(i))$  and  $\varphi(x) = (\varphi_1(x), \dots, \varphi_m(x))$ . Consequently, with  $\tilde{\Psi}_i := (v_k(i)/\sqrt{\tilde{\lambda}_k})_{k=1}^m$  (the same angle as  $\Psi_i$ , by invariances (i), (ii), (iv)) and provided  $\delta_n \leq \mu_1/2$ ,

$$\max_{i \leq n} \|Q_n \tilde{\Psi}_i - F(x_i)\| \leq E_n := \frac{\epsilon_n}{\sqrt{\mu_1}} + \frac{2\delta_n}{\mu_1^{3/2}} \sqrt{m} (\Phi_m + \epsilon_n) \rightarrow 0.$$

In particular the graph radius inherits the continuum band:  $\rho^- - E_n \leq \|\tilde{\Psi}_i\| \leq \rho^+ + E_n$  for every node  $i$ .

*Proof. (a) Spectral consistency.* For the unnormalized and random-walk graph Laplacians, [SM5] give, with probability tending to one, eigenvalue convergence  $|\tilde{\lambda}_k - \mu_k| \leq \delta_n$  and the gap stability that makes “fixed  $m$  at a spectral gap” well-posed, while [SM6] supplies the eigenvector consistency in  $\infty$ -norm with explicit rates (and an approximate  $C^{0,1}$ /gradient rate  $\epsilon'_n \rightarrow 0$  for the eigenvectors themselves, used for the cutoff in the remark below), stated per eigenvalue cluster. We make the block-orthogonal alignment explicit, since [SM6] (Theorem 2.6 with Remark 2.8) attaches to each discrete eigenvector  $v_k$  a *single* normalized limit eigenfunction  $\tilde{\varphi}_k$  (their sup-norm rate,  $\|v_k - \tilde{\varphi}_k\|_\infty \lesssim \mu_k \epsilon_n$ ), not an orthonormal eigenspace basis. Within a multiplet the partners  $\{\tilde{\varphi}_k\}$  need not be exactly orthonormal, but their *population* Gram matrix  $G = (\langle \tilde{\varphi}_k, \tilde{\varphi}_l \rangle_{L^2(\rho)})$  is  $I + O(\epsilon_n)$ , routed through the discrete vectors: the  $v_k$  are orthonormal in the empirical inner product ( $\frac{1}{n} \sum_i v_k(i) v_l(i) = \delta_{kl}$ ), each  $\tilde{\varphi}_k$  is sup-norm  $\epsilon_n$ -close to  $v_k$  on the sample, and empirical inner products concentrate on their  $L^2(\rho)$  values at the eigenvector-consistency rate, so  $\langle \tilde{\varphi}_k, \tilde{\varphi}_l \rangle_{L^2(\rho)} = \delta_{kl} + O(\epsilon_n)$ . Symmetric orthogonalization  $G^{-1/2} = I + O(\epsilon_n)$  therefore turns  $\{\tilde{\varphi}_k\}$  into an exactly  $L^2(\rho)$ -orthonormal basis of the limiting eigenspace at cost  $O(\epsilon_n)$  in sup-norm; the alignment of this basis to the reference eigenbasis  $\{\varphi_k\}$  is the *unique* block rotation  $Q_n \in O(m)$  (block-diagonal across multiplets, the orthogonal factor in the polar decomposition of the cross-Gram  $\langle \tilde{\varphi}_k, \varphi_l \rangle$ )—not  $G^{-1/2}$  itself, which is symmetric positive-definite. Each aligned discrete eigenvector is then  $(\epsilon_n/\sqrt{m})$ -close in sup-norm to its partner. (Multiplicity is exactly why the alignment is block-orthogonal rather than a per-mode sign choice; a simple eigenvalue is the  $1 \times 1$  block, where it *is* a sign choice.) Collecting the  $m$  per-mode bounds into the vector  $v(i)$  gives  $\|Q_n v(i) - \varphi(x_i)\| \leq \epsilon_n$ .

*(b) No normalization conversion is needed.* The calibration in (a) is for the random-walk eigenvectors, and by the exact per-node cancellation of Step 0(iv) the graph angular map is  $\hat{u}_i = \Psi_i^{\text{rw}} / \|\Psi_i^{\text{rw}}\|$  regardless of whether  $L_{\text{sym}}$  or  $L_{\text{rw}}$  is diagonalized: the algorithm’s embedding satisfies  $\Psi_i = \sqrt{k_i} \Psi_i^{\text{rw}}$  and the positive scalar  $\sqrt{k_i}$  cancels under row normalization. In particular no uniform degree concentration

is invoked and no bound on the degree ratio  $k_i/\bar{k}_n$  is required, so the transfer is valid under non-uniform sampling; the sampling density enters only by fixing the limiting (weighted) operator whose eigenfunctions  $\varphi_k$  appear in  $F$ , as in the Setting. (The similarity  $L_{\text{rw}} = D^{-1/2}L_{\text{sym}}D^{1/2}$  that underlies  $v_k^{\text{sym}} = D^{1/2}v_k^{\text{rw}}$  is exact, so the shared eigenvalue  $\lambda_k$  used throughout is unaffected.)

(c) *Bookkeeping.* Let  $W_\mu = \text{diag}(\mu_k^{-1/2})$  and  $\widetilde{W} = \text{diag}(\tilde{\lambda}_k^{-1/2})$ , so  $F(x) = W_\mu \varphi(x)$  and  $\tilde{\Psi}_i = \widetilde{W} v(i)$ . Since  $W_\mu$  is a scalar multiple of the identity on each multiplet and  $Q_n$  is block-diagonal across the multiplets,  $Q_n W_\mu = W_\mu Q_n$ . Hence

$$Q_n \tilde{\Psi}_i - F(x_i) = Q_n (\widetilde{W} - W_\mu) v(i) + W_\mu (Q_n v(i) - \varphi(x_i)).$$

For the first term,  $\delta_n \leq \mu_1/2$  gives  $\tilde{\lambda}_k \geq \mu_1/2$ , so

$$\|\widetilde{W} - W_\mu\|_{\text{op}} = \max_{k \leq m} \frac{|\mu_k - \tilde{\lambda}_k|}{\sqrt{\tilde{\lambda}_k} \mu_k (\sqrt{\tilde{\lambda}_k} + \sqrt{\mu_k})} \leq \frac{2\delta_n}{\mu_1^{3/2}},$$

while  $\|v(i)\| \leq \|Q_n v(i) - \varphi(x_i)\| + \|\varphi(x_i)\| \leq \epsilon_n + \sqrt{m} \Phi_m \leq \sqrt{m} (\Phi_m + \epsilon_n)$ . For the second term,  $\|W_\mu\|_{\text{op}} = \mu_1^{-1/2}$ . Adding the two bounds gives  $E_n$ . The radius statement is the reverse triangle inequality  $|\|\tilde{\Psi}_i\| - R(x_i)| = |\|Q_n \tilde{\Psi}_i\| - \|F(x_i)\|| \leq E_n$  together with  $\rho^- \leq R \leq \rho^+$  from (H2).  $\square$

*General filters (proof of the calibration claim in Corollary 3.9).* The proof above uses only two properties of the weight matrix  $W_\mu = \text{diag}(\mu_k^{-1/2})$ : it is constant on multiplets (hence commutes with the block-orthogonal  $Q_n$ ), and it is Lipschitz in the eigenvalue on the retained window. Both hold for any positive Lipschitz filter  $h$ : replacing  $W_\mu$  by  $H := \text{diag}(h(\mu_k))$  and  $\widetilde{W}$  by  $\text{diag}(h(\tilde{\lambda}_k))$ , step (c) goes through verbatim with  $\|\text{diag}(h(\tilde{\lambda}_k)) - H\|_{\text{op}} \leq L_h \delta_n$  and  $\|H\|_{\text{op}} \leq H_m$ , giving  $E_{n,h} = H_m \epsilon_n + L_h \delta_n \sqrt{m} (\Phi_m + \epsilon_n)$  as claimed; the commute filter  $h(\lambda) = \lambda^{-1/2}$  (with  $L_h \leq 2\mu_1^{-3/2}$  on  $[\mu_1/2, \infty)$  and  $H_m = \mu_1^{-1/2}$ ) recovers the  $E_n$  displayed above.

*Conditional structure, and the bridge to uniformity in  $m$ .* Step 3 is *external*: it composes the named rate theorems [SM5, SM6] into the calibration, and the transfer of the continuum bi-Lipschitz constants to the graph is then the elementary node-uniform perturbation of Step 4 (radius floor  $\|\tilde{\Psi}_i\| \geq \rho^- - E_n$ ; angular error  $\leq 2E_n/\rho^-$  by Lemma SM1.2; cutoff  $\theta_n \asymp E_n/a$ ). Consequently Theorem 3.10 is **unconditional in  $m$  for each fixed  $m$** , resting only on this standard sup-norm spectral-transfer lemma—no self-contained spectral analysis is re-derived here. The one  $m$ -dependent seam is deliberate and worth stating plainly: the CGLewicka constants, hence  $\epsilon_n$  and  $\epsilon'_n$ , depend on the retained eigenfunctions' Lipschitz norms and on the spectral gap at  $\mu_m$ , so the *rate* at which  $\theta_n \rightarrow 0$  degrades as  $m$  grows. Making the transfer *uniform* in  $m$ —and with it (H1)—is exactly the frontier of Conjecture 3.4; we delimit it sharply in the discussion after Proposition 3.17, and it is filter-specific, since the  $C^{0,1}$  rate  $\epsilon'_n$  is what shrinks  $\theta_n$  toward the connectivity scale (below which the twin obstruction lives).

#### Step 4: assembly.

*Proof of Theorem 3.10.* Work on the event  $\mathcal{A}_n$  of Lemma SM1.3, with  $n$  large enough that  $E_n < \rho^-/2$  (so  $\tilde{\Psi}_i \neq 0$  and every  $\hat{u}_i$  is defined). By invariance (iii) we may compare angles through the calibrated embedding: writing  $\hat{q}_i := Q_n \tilde{\Psi}_i / \|Q_n \tilde{\Psi}_i\| =$

162  $Q_n \hat{u}_i$ , Lemma SM1.2 with  $u = Q_n \tilde{\Psi}_i$ ,  $w = F(x_i)$ , and  $\|F(x_i)\| \geq \rho^-$  gives

$$163 \quad \|\hat{q}_i - N(x_i)\| \leq \frac{2E_n}{\rho^-} \quad \text{for every node } i.$$

164 Set  $\theta_n := 8E_n/(a\rho^-) \rightarrow 0$ , and fix a sampled pair with  $\theta_n \leq d := d_M(x_i, x_j) \leq r_0$ . The  
 165 pairwise angular error is at most  $4E_n/\rho^- = (a/2)\theta_n \leq (a/2)d$ , so by Lemma SM1.1,  
 166 the triangle inequality, and  $\|\hat{u}_i - \hat{u}_j\| = \|\hat{q}_i - \hat{q}_j\|$ ,

$$167 \quad \|\hat{u}_i - \hat{u}_j\| \geq ad - \frac{a}{2}d = \frac{a}{2}d = Ad, \quad \|\hat{u}_i - \hat{u}_j\| \leq bd + \frac{a}{2}d \leq 2bd = Bd,$$

168 the last inequality using  $a \leq b$  (Step 1). Both bounds hold simultaneously for *all* such  
 169 pairs because the calibration of Lemma SM1.3 is uniform over nodes, and  $\mathbb{P}(\mathcal{A}_n) \rightarrow 1$ .  $\square$

170 **Step 5: from local to global (proof of Corollary 3.11).**

171 *Proof of Corollary 3.11.* The map  $(x, y) \mapsto \|N(x) - N(y)\|$  is continuous ( $F$  is  
 172 continuous and  $R \geq \rho^- > 0$ ) and, by (H3), strictly positive on the compact set  
 173  $\{(x, y) \in M \times M : d_M(x, y) \geq r_0\}$ ; its minimum  $c_0$  is therefore attained and posi-  
 174 tive. Work on the event of Lemma SM1.3 with  $n$  large enough that  $E_n < \rho^-/2$  and  
 175  $4E_n/\rho^- \leq c_0/2$ . For sampled pairs with  $\theta_n \leq d_M \leq r_0$ , Theorem 3.10 gives the  
 176 bounds with  $(A, B)$ , hence with  $(A_g, B_g)$ . For sampled pairs with  $d := d_M > r_0$ : the  
 177 calibration bound  $\|\hat{q}_i - N(x_i)\| \leq 2E_n/\rho^-$  of Step 4 gives

$$178 \quad \|\hat{u}_i - \hat{u}_j\| \geq \|N(x_i) - N(x_j)\| - \frac{4E_n}{\rho^-} \geq c_0 - \frac{c_0}{2} = \frac{c_0}{2} \geq \frac{c_0}{2 \operatorname{diam} M} d \geq A_g d,$$

179 while unit vectors satisfy  $\|\hat{u}_i - \hat{u}_j\| \leq 2 \leq (2/r_0)d \leq B_g d$ .  $\square$

180 **Step 6: the flat torus (proof of Corollary 3.12).**

181 *Proof of Corollary 3.12.* On  $M = \mathbb{R}^2/(2\pi\mathbb{Z})^2$  with volume normalized to 1,  
 182 the Laplace–Beltrami eigenvalues are  $|k|^2$ ,  $k \in \mathbb{Z}^2$ : the first nontrivial eigen-  
 183 value  $\mu = 1$  has multiplicity four, spanned by the  $L^2$ -orthonormal functions  
 184  $\sqrt{2}\cos x$ ,  $\sqrt{2}\sin x$ ,  $\sqrt{2}\cos y$ ,  $\sqrt{2}\sin y$ , and the next eigenvalue is 2—so  $m = 4$   
 185 exhausts the first multiplet and sits at a spectral gap. The truncated eigenmap  
 186 (weights  $1/\sqrt{\mu_k} = 1$ ) is the Clifford embedding  $F(x, y) = \sqrt{2}(\cos x, \sin x, \cos y, \sin y)$ .  
 187 (H2).  $\|F\|^2 = 2(\cos^2 x + \sin^2 x) + 2(\cos^2 y + \sin^2 y) = 4$ : the radius is constant,  
 188  $R \equiv 2$ , so  $\Lambda_0 = 0 < 1/L_0$  and  $\rho^- = \rho^+ = 2$ .

189 (H1), globally. Using  $(\cos a - \cos b)^2 + (\sin a - \sin b)^2 = 4\sin^2(\frac{a-b}{2})$ ,

$$190 \quad \|F(p) - F(q)\|^2 = 8\left[\sin^2 \frac{\Delta x}{2} + \sin^2 \frac{\Delta y}{2}\right], \quad \Delta x, \Delta y \in [-\pi, \pi],$$

191 while  $d_M(p, q)^2 = \Delta x^2 + \Delta y^2$ . The elementary bounds  $\frac{2}{\pi}|\delta| \leq 2|\sin(\delta/2)| \leq |\delta|$  on  
 192  $[-\pi, \pi]$  give

$$193 \quad \frac{2\sqrt{2}}{\pi} d_M(p, q) \leq \|F(p) - F(q)\| \leq \sqrt{2} d_M(p, q) \quad \text{for all } p, q,$$

194 so (H1) holds with  $L_0 = \sqrt{2}$  (indeed  $L_0^{-1} = 1/\sqrt{2} \leq 2\sqrt{2}/\pi$ ) and  $r_0 = \operatorname{diam} M = \pi\sqrt{2}$ .

195 (H3).  $N = F/2$ ; since the radius is constant,  $N(p) = N(q)$  iff  $F(p) = F(q)$  iff  
 196  $e^{ix_p} = e^{ix_q}$  and  $e^{iy_p} = e^{iy_q}$  iff  $p = q$ .

197 Because  $r_0 = \operatorname{diam} M$ , Theorem 3.10 alone is already global, with  $A =$   
 198  $\sqrt{L_0^{-2}/(2\rho^+)} = 1/(4\sqrt{2})$  and  $B = 2L_0/\rho^- = \sqrt{2}$ . The sharp continuum distortion of

$N$  is the exact ratio range above:  $\|N(p) - N(q)\|/d_M(p, q) \in [\sqrt{2}/\pi, 1/\sqrt{2}]$ , the ratio of whose extremes is  $\pi/2$ , attained along a coordinate direction at antipodal ( $|\delta| = \pi$ ) and infinitesimal separations respectively. Basis-independence: any  $L^2$ -orthonormal basis of the  $\mu = 1$  eigenspace equals  $Q$  applied to the Clifford one for some  $Q \in O(4)$ , and angular pairwise distances are  $Q$ -invariant (Step 0, invariance (iii)).  $\square$

**Step 7: the small-ball ratio law (used by Proposition 3.14).** Let  $D_1, D_2$  be i.i.d. with the small-ball distance law  $F(r) = (r/r_0)^d$  on  $[0, r_0]$  (density  $\propto r^{d-1}$ , the flat approximation to pair distances within a geodesic ball). For  $\kappa \geq 1$ ,

$$\mathbb{P}(D_2 \geq \kappa D_1) = \int_0^{r_0} F\left(\frac{y}{\kappa}\right) dF(y) = \frac{1}{\kappa^d} \int_0^{r_0} \left(\frac{y}{r_0}\right)^d \frac{dy^{d-1}}{r_0^d} dy = \frac{1}{2\kappa^d},$$

so by symmetry  $\mathbb{P}(\max/\min \geq \kappa) = \kappa^{-d}$  and  $\mu(\kappa) = 1 - \kappa^{-d}$  exactly. The rank certificate of Proposition 3.14 is thus  $\tau \geq 2\kappa^{-d} - 1$ ,  $\rho_S \geq 3\kappa^{-d} - 2$ : exponential decay of the worst case in the intrinsic dimension.

**Why the resolution cutoff is necessary.** The cutoff  $\theta_n$  is forced by the statement's generality, not by the method. In a hard-threshold  $\varepsilon$ -graph, two sample points at distance below the bandwidth can be *combinatorial twins*—identical neighbourhoods, hence identical rows of  $A$  and equal degrees. For twins  $i, j$  the difference  $e_i - e_j$  is itself an eigenvector of  $L_{\text{sym}}$ —at eigenvalue 1 for non-adjacent twins and  $1 + 1/k$  for adjacent ones—so every eigenvector at any *other* eigenvalue is orthogonal to it and has identical entries at  $i$  and  $j$ . In particular every low mode does, hence  $\hat{u}_i = \hat{u}_j$  exactly while  $d_M(x_i, x_j) > 0$ : no lower bound  $A d_M$  can hold below the connectivity scale. (“Identical in every eigenvector” would be literally false—the localized twin-difference mode itself separates them—but that mode sits at the top of the spectrum, far above anything the angular map retains.) What is improvable is the *size* of  $\theta_n$ : our proof takes it at the sup-norm consistency rate  $E_n$ , much larger than the bandwidth; a  $C^1$  (gradient) consistency argument—the Lipschitz regularity theory for graph-Laplacian eigenfunctions of [SM6] is the natural ingredient—would control eigenvector *differences* at small separations directly and could shrink  $\theta_n$  toward the connectivity scale, below which the twin obstruction is final.

Two further remarks. (1) Every constant in  $E_n$  depends on the fixed mode-count  $m$  (through  $\sqrt{m}$ ,  $\Phi_m$ ,  $\mu_1$ , and the gap  $\mu_{m+1} - \mu_m$ ); nothing here is claimed uniform in  $m$ . The uniformity in  $m$  observed empirically (§5.2) corresponds to the uniformity of (H1) [SM1, SM2] and is exactly what separates Theorem 3.10 from the full Conjecture 3.4; Appendix SM2 settles its continuum side by filter (uniform for heat, scale-dependent for commute). (2) The theorem is local ( $d_M \leq r_0$ ) because (H1) is; the global statement needs precisely the injectivity hypothesis (H3), which Corollary 3.11 (Step 5) carries out—and on the flat torus (H1) is itself global, so Corollary 3.12 (Step 6) needs none of (H1)–(H3).

**SM2. Uniformity in the truncation.** This appendix proves Theorem 3.15 and Proposition 3.16, and records the growing-window graph-transfer conditions. Throughout, volume is normalized to 1,  $p_t$  is the heat kernel of  $M$ , and cutoffs are multiplet-closed.

**Proof of Theorem 3.15.** Let  $F_t(x) = (e^{-t\mu_k/2} \varphi_k(x))_{k \geq 1} \in \ell^2$  be the *full* heat embedding with the constant mode removed; its kernel is

$$K_t(x, y) := \langle F_t(x), F_t(y) \rangle = p_t(x, y) - 1.$$

243 *Step 1:  $N_t$  is a uniformly noncollapsing immersion for small  $t$ .* For a unit tangent  
 244 vector  $v \in T_x M$ ,

$$245 \quad \|dF_t(v)\|^2 = \partial_x^v \partial_y^v K_t(x, y)|_{y=x}, \quad dR_t(v) = \frac{\partial_v K_t(x, x)}{2R_t(x)}, \quad R_t^2(x) = p_t(x, x) - 1.$$

246 By the Minakshisundaram–Pleijel expansion (cf. [SM3]),  $p_t(x, x) = (4\pi t)^{-d/2}(1 +$   
 247  $t \operatorname{scal}(x)/6 + O(t^2))$  and  $\partial_x^v \partial_y^v p_t|_{y=x} = \frac{1}{2t}(4\pi t)^{-d/2}(1 + O(t))$ , both uniformly on com-  
 248 pact  $M$ . The only  $x$ -dependence of the diagonal at this order is through the curvature  
 249 term, so  $\partial_v p_t(x, x) = (4\pi t)^{-d/2} O(t)$ , whence  $dR_t(v)^2 = (4\pi t)^{-d/2} O(t^2) \cdot O(1)$ , which  
 250 is  $o$  of the derivative energy  $\asymp t^{-1}(4\pi t)^{-d/2}$ . Therefore there is  $t_0(M) > 0$  such that  
 251 for  $t \leq t_0$  the tangential square  $T_t(x, v)^2 := \|dF_t(v)\|^2 - dR_t(v)^2$  obeys

$$252 \quad T_t(x, v)^2 \geq c t^{-d/2-1}, \quad R_t(x)^2 \asymp t^{-d/2}, \quad \text{hence} \quad \|dN_t(v)\|^2 = \frac{T_t(x, v)^2}{R_t(x)^2} \geq \frac{c'}{t},$$

253 uniformly over the unit tangent bundle (the last identity is Proposition 3.7).

254 *Step 2: the truncation tail vanishes in  $C^2$ .* Eigenfunction Hölder/derivative norms  
 255 grow only polynomially,  $\|\varphi_k\|_{C^r} \leq C \mu_k^{(d-1)/4+r/2}$  [SM9], while the heat weights decay  
 256 exponentially; hence  $\|F_t - F_{t,m}\|_{C^2} \rightarrow 0$  as  $m \rightarrow \infty$  (tail sums  $\sum_{k>m} e^{-t\mu_k} \mu_k^{c(d,r)} \rightarrow 0$   
 257 by Weyl's law). Consequently  $T_{t,m} \rightarrow T_t$ ,  $R_{t,m} \rightarrow R_t$ , and the  $C^2$ -norms of  $N_{t,m}$   
 258 converge, all uniformly; choose  $m_0(t)$  so that for  $m \geq m_0(t)$ ,  $T_{t,m}^2 \geq \frac{1}{2} c t^{-d/2-1}$ ,  
 259  $R_{t,m} \asymp t^{-d/4}$ , and  $\sup_{m \geq m_0} \|N_{t,m}\|_{C^2} =: C_2(t) < \infty$ .

260 *Step 3: from uniform immersion to uniform bi-Lipschitz.* Let  $a'_t :=$   
 261  $\inf_{m \geq m_0} \inf_{x, |v|=1} \|dN_{t,m}(v)\| > 0$  from Steps 1–2. For  $x, y$  with  $d := d_M(x, y) \leq$   
 262  $r_t := a'_t/(2C_2(t))$ , Taylor along a minimizing geodesic  $\gamma$  gives

$$263 \quad \|N_{t,m}(x) - N_{t,m}(y)\| \geq \|dN_{t,m}(\gamma'(0))\| d - \frac{1}{2} C_2(t) d^2 \geq \left(a'_t - \frac{1}{2} C_2(t) r_t\right) d \geq \frac{3a'_t}{4} d,$$

264 and the mean-value bound gives  $\|N_{t,m}(x) - N_{t,m}(y)\| \leq b_t d$  with  $b_t :=$   
 265  $\sup_{m \geq m_0} \operatorname{Lip}(N_{t,m}) < \infty$ . All constants depend on  $t$  and  $M$  only—not on  
 266  $m$ .

267 **LEMMA SM2.1** (Differentiated pointwise Weyl law). *Let  $(M, g)$  be a smooth*  
 268 *closed Riemannian  $d$ -manifold with orthonormal Laplace eigenfunctions  $\varphi_k$ , eigenval-*  
 269 *ues  $\mu_k$ . There are constants  $c_d, c'_d > 0$  depending only on  $(M, g)$  such that, uniformly*  
 270 *in  $x \in M$ ,*

$$271 \quad \sum_{0 < \mu_k \leq \Lambda} \varphi_k(x)^2 = c_d \Lambda^{d/2} (1 + O(\Lambda^{-1/2})), \quad \sum_{0 < \mu_k \leq \Lambda} \|\nabla \varphi_k(x)\|^2 = c'_d \Lambda^{(d+2)/2} (1 + O(\Lambda^{-1/2})),$$

272 *with spatially constant leading coefficients.*

273 *Proof.* Both are on-diagonal statements about the spectral projector  $\Pi_\lambda(x, y) =$   
 274  $\sum_{\sqrt{\mu_k} \leq \lambda} \varphi_k(x) \varphi_k(y)$  at frequency  $\lambda = \sqrt{\Lambda}$ : the first is Hörmander's pointwise Weyl  
 275 law with sharp remainder [SM9],  $\Pi_\lambda(x, x) = \tilde{c}_d \lambda^d + O(\lambda^{d-1})$ ; the second is the mixed-  
 276 derivative case  $\partial_x \partial_y \Pi_\lambda(x, y)|_{y=x} = \tilde{c}'_d \lambda^{d+2} + O(\lambda^{d+1})$ , the uniform derivative estimates  
 277 for the spectral function on a closed manifold of [SM4] (one power of  $\lambda$  below the main  
 278 term, uniformly on  $M$ ; no non-self-focal restriction, which the finer scaling limit of  
 279 [SM7] would carry). Substituting  $\lambda = \sqrt{\Lambda}$  gives the stated relative remainders.  $\square$

**Proof of Proposition 3.16.** Let  $E_\Lambda(x, y) = \sum_{\mu_k \leq \Lambda} \varphi_k(x) \varphi_k(y)$  be the spectral function. The pointwise Weyl law with uniform remainder [SM9] gives  $E_\Lambda(x, x) = C_d \Lambda^{d/2} + O(\Lambda^{(d-1)/2})$  with spatially constant leading coefficient, and its differentiated form (the uniform derivative-Weyl estimates of [SM4]; equivalently Lemma SM2.1) gives, uniformly in  $x$  and unit  $v$ ,

$$\partial_x^v \partial_y^v E_\Lambda(x, y) \Big|_{y=x} = C'_d \Lambda^{d/2+1} (1 + o(1)).$$

Abel summation in the cutoff converts these into the commute-weighted sums:

$$\|dF_\Lambda(v)\|^2 = \sum_{0 < \mu_k \leq \Lambda} \mu_k^{-1} (\partial_v \varphi_k(x))^2 = c_d \Lambda^{d/2} (1 + o(1)),$$

$$G_\Lambda(x, x) := R_\Lambda^2(x) = \sum_{0 < \mu_k \leq \Lambda} \mu_k^{-1} \varphi_k(x)^2 = \begin{cases} C_d \Lambda^{d/2-1} (1 + o(1)), & d > 2, \\ C_2 \log \Lambda (1 + o(1)), & d = 2, \end{cases}$$

again uniformly in  $x$  (Lemma SM2.1 in the commute-weighted normalization; Abel summation in the cutoff converts the on-diagonal projector asymptotics into the  $\mu^{-1}$ -weighted sums), with spatially constant leading coefficients. Because the leading coefficient of  $G_\Lambda(x, x)$  is spatially constant, its spatial derivative has no leading term, and the differentiated remainder estimates of [SM4] bound what remains,  $\partial_v G_\Lambda(x, x) = o(\Lambda^{(d-1)/2})$  uniformly—*weaker* than the naive  $o(\Lambda^{d/2-1})$  and all that is needed—so the radial term  $dR_\Lambda(v)^2 = (\partial_v G_\Lambda(x, x))^2 / (4G_\Lambda(x, x)) = o(\Lambda^{d/2})$  is lower order than the derivative energy. Hence  $T_\Lambda(x, v)^2 \geq c \Lambda^{d/2}$  for  $\Lambda \geq \Lambda_0$ , uniformly, which is the display; dividing by  $R_\Lambda^2$  gives  $\|dN_\Lambda\| \gtrsim \Lambda^{1/2}$  ( $d > 2$ ) and  $\sqrt{\Lambda / \log \Lambda}$  ( $d = 2$ ).

*The  $S^1$  computation, and the failure of fixed-scale uniformity.* On  $S^1$  ( $\mu = k^2$ , whole eigenspaces, weights  $\mu^{-1/2} = 1/k$ ),

$$F_N(x) = \left( \frac{\cos kx}{k}, \frac{\sin kx}{k} \right)_{k=1}^N : \quad R_N^2 = \sum_{k \leq N} \frac{1}{k^2} \text{ (constant, bounded),} \quad \|dF_N\|^2 = \sum_{k \leq N} 1 = N,$$

so  $\|dN_N\| = \|dF_N\|/R_N \asymp \sqrt{N}$  (consistent with the general formula:  $\Lambda = N^2$ ,  $d = 1$ ,  $T_\Lambda/R \asymp \Lambda^{1/4}$ ). A pair at distance  $\sim 1/N$  is stretched by  $\sim \sqrt{N}$ , so any upper Lipschitz constant valid on a *fixed* ball  $\{d \leq r_0\}$  must grow like  $\sqrt{m}$ : no  $m$ -independent fixed-scale bi-Lipschitz theorem can hold for the commute filter, on any manifold (the same growth holds in all  $d$ ). The lower bound at fixed scale, by contrast, survives on  $S^1$  (project onto the first multiplet, whose weight is  $m$ -independent while  $R_N$  is bounded)—the failure is one-sided and lives below the wavelength  $\Lambda^{-1/2}$ .

*Remark SM2.2 (Cutoff-insensitivity).* The conclusion of Proposition 3.16 does not depend on the sharp spectral cutoff: replacing  $\sum_{\mu_k \leq \Lambda}$  by a smoothed Littlewood–Paley weight  $\chi(\mu_k/\Lambda)$  yields the same growth orders with full asymptotic expansions (smoothed spectral sums admit wave-kernel expansions without Tauberian remainders), and on  $S^1$  the sharp-cutoff computation is exact with no Weyl input at all. The proposition’s content—no  $m$ -independent fixed-scale upper Lipschitz constant for the commute filter—is cutoff-independent.

### A uniform lower angular bound on flat tori.

**PROPOSITION SM2.3** (Uniform lower angular bound on flat tori). *Let  $T = \mathbb{R}^d/\Gamma$  be a fixed flat torus with dual lattice  $\Gamma^*$ . For a multiplet-closed spectral cutoff  $\Lambda$ , set*

$$F_\Lambda(x) = \left( \frac{e^{i\langle \xi, x \rangle}}{|\xi|} \right)_{\xi \in \Gamma^* \setminus \{0\}, |\xi|^2 \leq \Lambda}, \quad N_\Lambda(x) = \frac{F_\Lambda(x)}{\|F_\Lambda(x)\|}$$

(equivalently, pair  $\xi$  with  $-\xi$  and use real sine-cosine coordinates). There exist  $\Lambda_0(T) < \infty$  and  $c_T > 0$  such that

$$\|N_\Lambda(x) - N_\Lambda(y)\| \geq c_T d_T(x, y)$$

for every multiplet-closed  $\Lambda \geq \Lambda_0(T)$  and every  $x, y \in T$ ; the lower angular Lipschitz constant is thus uniform in the mode count  $m = m(\Lambda)$ . For the standard torus  $\mathbb{T}^d = \mathbb{R}^d / (2\pi\mathbb{Z})^d$  one may take  $\Lambda_0 = 1$ .

*Proof.* We work first on the standard torus; the general lattice is treated at the end. Write  $R = \sqrt{\Lambda}$  and  $\mathcal{K}_R = \{k \in \mathbb{Z}^d \setminus \{0\} : |k| \leq R\}$ . Because the cutoff contains complete sine-cosine pairs,  $S_R := \|F_R(x)\|^2 = \sum_{k \in \mathcal{K}_R} |k|^{-2}$  is independent of  $x$ . If  $z = x - y$  is represented in  $[-\pi, \pi]^d$  then  $d_{\mathbb{T}^d}(x, y) = |z|$  and

$$(SM2.1) \quad \|N_R(x) - N_R(y)\|^2 = \frac{2A_R(z)}{S_R}, \quad A_R(z) := \sum_{k \in \mathcal{K}_R} \frac{1 - \cos(k \cdot z)}{|k|^2}.$$

We prove  $A_R(z) \geq c_d |z|^2 S_R$  with  $c_d > 0$  independent of  $R$ .

*A Fourier-block estimate.* For  $q \geq 2$  and  $t \in [-\pi, \pi]$  set  $a_q(t) := \frac{1}{q} \sum_{n=q}^{2q-1} \cos(nt)$ . There is an absolute  $c > 0$  with

$$(SM2.2) \quad 1 - |a_q(t)| \geq c \min\{1, q^2 t^2\}.$$

Indeed, if  $U$  is uniform on  $\{q, \dots, 2q-1\}$  and  $\phi_q(t) = \mathbb{E} e^{iUt}$ , then  $|a_q(t)| \leq |\phi_q(t)|$ . When  $q|t| \leq 1$ , for an independent copy  $U'$ ,

$$1 - |\phi_q(t)| \geq \frac{1}{2} (1 - |\phi_q(t)|^2) = \frac{1}{2} \mathbb{E}[1 - \cos((U - U')t)] \geq c q^2 t^2,$$

using  $1 - \cos s \geq cs^2$  for  $|s| \leq 1$  and  $\mathbb{E}(U - U')^2 \asymp q^2$ . When  $q|t| \geq 1$ , the exact formula  $|\phi_q(t)| = |\sin(qt/2)| / (q|\sin(t/2)|)$  is bounded by a universal  $\rho < 1$ : for bounded  $q$  the ratio is  $< 1$  on the compact locus  $q|t| \geq 1$ , and as  $q \rightarrow \infty$  it tends to 0 unless  $t \rightarrow 0$ , where with  $s = qt/2 \geq \frac{1}{2}$  it equals  $|\sin s|/s < 1$ . This gives (SM2.2).

*Lattice blocks.* Choose  $j$  with  $|z_j| = \|z\|_\infty$ . For dyadic  $q$  put  $B_{q,j} = \{k \in \mathbb{Z}^d : q \leq |k_j| < 2q, |k_\ell| < q (\ell \neq j)\}$ , disjoint across dyadic  $q$ , with  $|B_{q,j}| \asymp_d q^d$  and  $|k|^2 \leq (d+3)q^2$  on  $B_{q,j}$ . The block average of  $\cos(k \cdot z)$  factors as  $a_q(z_j) \prod_{\ell \neq j} b_q(z_\ell)$  with each normalized symmetric Dirichlet sum  $|b_q| \leq 1$ , so by (SM2.2)

$$\frac{1}{|B_{q,j}|} \sum_{k \in B_{q,j}} (1 - \cos(k \cdot z)) \geq 1 - |a_q(z_j)| \geq c \min\{1, q^2 z_j^2\}.$$

With  $|k|^2 \leq (d+3)q^2$ ,  $|B_{q,j}| \asymp q^d$ , and  $\min\{1, q^2 z_j^2\} \geq |z|^2 / (d\pi^2)$  (from  $z_j^2 \geq |z|^2 / d$  and  $|z| \leq \sqrt{d}\pi$ ),

$$\sum_{k \in B_{q,j}} \frac{1 - \cos(k \cdot z)}{|k|^2} \geq c_d q^{d-2} |z|^2.$$

Every block with  $q \leq R/\sqrt{d+3}$  lies in  $\mathcal{K}_R$ ; summing over dyadic  $q$  in that range,  $A_R(z) \geq c_d |z|^2 \sum_{q \text{ dyadic}, 2 \leq q \leq R/\sqrt{d+3}} q^{d-2}$ , while a dyadic decomposition of the lattice ball gives  $S_R \leq C_d (1 + \sum_{q \text{ dyadic}, q \leq R} q^{d-2})$ . The two sums are comparable for large  $R$ —bounded for  $d = 1$ ,  $\asymp \log R$  for  $d = 2$ ,  $\asymp R^{d-2}$  for  $d \geq 3$ —so  $A_R(z) \geq c_d |z|^2 S_R$ . For the finitely many smaller cutoffs the first multiplet already gives  $A_R(z) \geq \sum_j (1 -$

$\cos z_j) \geq c_d |z|^2$  with  $S_R$  bounded, so the bound holds for all  $R \geq 1$ . Substituting into (SM2.1),  $\|N_R(x) - N_R(y)\|^2 \geq c_d |z|^2 = c_d d_{\mathbb{T}^d}(x, y)^2$ .

For a general flat torus, a dual-lattice basis  $b_1, \dots, b_d$  and norm equivalence  $c_T |n| \leq |\xi(n)| \leq C_T |n|$  reduce the estimate to the standard case in angular coordinates  $\theta \in [-\pi, \pi]^d$ , with constants depending on  $T$ ; taking  $\Lambda_0(T)$  large enough to contain a dual basis handles the initial cutoffs.  $\square$

Together with Proposition 3.16 this locates the commute frontier exactly:  $c_T d_T(x, y) \leq \|N_m(x) - N_m(y)\|$  uniformly in  $m$ , while there is *no*  $m$ -uniform upper bound at a fixed spatial scale, since  $\|dN_m\|$  grows with the cutoff. The lower Lipschitz half of the metric conjecture is thus proved on the reference substrate family; only the scale-dependent upper half remains.

*Remark SM2.4* (Exact upper-Lipschitz divergence). The companion upper bound diverges, on  $\mathbb{T}^d$  exactly and for every  $d$ . Because  $S_R$  is independent of  $x$ , normalization produces no radial cancellation, so for a coordinate direction  $e_j$ ,

$$\|dN_R(e_j)\|^2 = \frac{1}{S_R} \sum_{k \in \mathcal{K}_R} \frac{k_j^2}{|k|^2} = \frac{|\mathcal{K}_R|}{d S_R},$$

using  $\sum_{j=1}^d k_j^2 / |k|^2 = 1$  at every lattice point. With  $|\mathcal{K}_R| \asymp R^d$  and the orders of  $S_R$  above,

$$\text{Lip}(N_R) \asymp \begin{cases} R^{1/2}, & d = 1, \\ R/\sqrt{\log R}, & d = 2, \\ R, & d \geq 3, \end{cases}$$

so no  $R$ -independent upper Lipschitz constant exists on any fixed geodesic ball. This is an exact obstruction, not a gap in technique, and it survives a global output rescaling  $N_R \mapsto c_R N_R$ : uniform upper control forces  $c_R \rightarrow 0$ , which collapses the fixed-diameter image ( $\text{diam} \leq 2c_R \rightarrow 0$ ) and destroys the lower bound. The commute map is therefore *uniformly co-Lipschitz with a quantitatively diverging upper Lipschitz constant*; genuine two-sided uniformity requires a faster filter—the heat weighting (Theorem 3.15), or amplitude  $|k|^{-s}$  with  $s > (d+2)/2$ , for which the numerator above converges while  $S_R$  tends to a positive constant.

**A spectral-filter phase diagram.** The commute weighting is one member of the fractional family

$$F_{\Lambda,s}(x) = (\mu_k^{-s/2} \varphi_k(x))_{0 < \mu_k \leq \Lambda}, \quad K_{\Lambda,s}(x, y) = \sum_{0 < \mu_k \leq \Lambda} \mu_k^{-s} \varphi_k(x) \varphi_k(y),$$

with commute at  $s = 1$ , the plain eigenmap at  $s = 0$ , and the heat weighting a rapidly decaying alternative. Weyl's law  $\mu_k \asymp k^{2/d}$  separates three regimes by three convergence thresholds.

PROPOSITION SM2.5 (Filter thresholds). As  $\Lambda \rightarrow \infty$ ,

$$\sum_k \mu_k^{-2s} < \infty \iff s > \frac{d}{4}, \quad \sum_k \mu_k^{-s} < \infty \iff s > \frac{d}{2}, \quad \sum_k \mu_k^{1-s} < \infty \iff s > \frac{d}{2} + 1,$$

controlling respectively: (i) the  $L^2(M \times M)$  (Hilbert–Schmidt) limit of the correction kernel  $K_{\Lambda,s}$ —hence stability of the rank information it carries; (ii) the trace  $\int_M \|F_{\Lambda,s}\|^2 = \sum_k \mu_k^{-s}$ , i.e. finiteness of the radius in the infinite embedding; and (iii)

the differential energy  $\int_M \|dF_{\Lambda,s}\|^2 = \sum_k \mu_k^{1-s}$ , i.e. a finite uniform upper Lipschitz constant.

*Proof.* Each equivalence is Weyl's law:  $\mu_k \asymp k^{2/d}$  gives  $\mu_k^{-a} \asymp k^{-2a/d}$ , and  $\sum_k k^{-2a/d} < \infty \iff 2a/d > 1$ ; take  $a = 2s$ ,  $s, s-1$ . For (i), orthonormality gives  $\|K_{\Lambda,s}\|_{L^2(M \times M)}^2 = \sum_{\mu_k \leq \Lambda} \mu_k^{-2s}$ , Cauchy in  $\Lambda$  iff the sum converges. Claims (ii)–(iii) integrate  $\|F\|^2$  and  $\|dF\|^2$  over  $M$ .  $\square$

For the commute filter  $s = 1$  the thresholds read  $d < 4$ ,  $d < 2$ , and  $d < 0$ :

| Property                           | Convergent sum    | Commute ( $s = 1$ )                 |
|------------------------------------|-------------------|-------------------------------------|
| Rank kernel Hilbert–Schmidt stable | $\sum \mu_k^{-2}$ | holds for $d \leq 3$                |
| Radius finite in the limit         | $\sum \mu_k^{-1}$ | only $d = 1$ (diverges $d \geq 2$ ) |
| Uniform upper Lipschitz bound      | $\sum \mu_k^0$    | fails $\forall d \geq 1$            |

This resolves three observations at once: rank stability can survive in  $d = 1, 2, 3$  (the empirical “flat in  $m$ ”); the radius changes character at  $d = 2$  (Corollary 3.1); and two-sided uniform metric control fails in every positive dimension (Proposition 3.16, and the exact torus divergence of the remark above). The commute filter sits at the *critical* exponent for the rank kernel, whose stability boundary is the dimension  $d = 4$ —a sharper structural prediction than the five-family linear trend of Paper I.b. A filter with  $s > d/2 + 1$  (the heat weighting, or amplitude  $\mu_k^{-s/2}$  with  $s > d/2 + 1$ ) restores a uniform two-sided bound.

**A rank-limit theorem on flat tori ( $d \leq 3$ ).** On a flat torus the diagonal kernel  $S_\Lambda = \|F_\Lambda(x)\|^2$  is independent of  $x$ , so the angular distance is an exact decreasing function of a single scalar:

$$\|N_\Lambda(x) - N_\Lambda(y)\|^2 = 2 - \frac{2G_\Lambda(x-y)}{S_\Lambda}, \quad G_\Lambda(z) = \sum_{0 < |k|^2 \leq \Lambda} \frac{\cos(k \cdot z)}{|k|^2}.$$

Because  $S_\Lambda$  does not depend on the pair, the angular distances carry exactly the reverse ranking of  $G_\Lambda(x-y)$ —so although every distance concentrates at  $\sqrt{2}$  as  $\Lambda \rightarrow \infty$  (the radius diverges,  $S_\Lambda \rightarrow \infty$  for  $d \geq 2$ ), the *ordering* that rank statistics read need not degenerate.

**PROPOSITION SM2.6 (Rank stability on flat tori).** *Let  $d \leq 3$ . Then  $\sum_{k \neq 0} |k|^{-4} < \infty$ , so  $G_\Lambda \rightarrow G$  in  $L^2(\mathbb{T}^d)$ , where  $G$  is the zero-mean periodic Green's function. If  $(X, Y)$  is drawn from an absolutely continuous pair law (uniform sampling suffices) whose induced law of  $G(X - Y)$  is atomless, then the population Spearman and Kendall correlations between  $d_{\mathbb{T}^d}(X, Y)$  and  $\|N_\Lambda(X) - N_\Lambda(Y)\|$  converge to those of  $(d_{\mathbb{T}^d}(X, Y), -G(X - Y))$ . On  $\mathbb{S}^1$  the periodic Green's function is strictly monotone in geodesic separation, so the limiting rank correlation is exactly 1.*

*Proof.*  $G$  has Fourier coefficients  $|k|^{-2}$ , so  $\|G_\Lambda - G\|_{L^2}^2 = \sum_{|k|^2 > \Lambda} |k|^{-4} \rightarrow 0$  exactly when  $\sum_k |k|^{-4} < \infty$ , i.e.  $4 > d$ . Hence  $G_\Lambda(X - Y) \rightarrow G(X - Y)$  in probability along the pair law. The angular distance is a fixed continuous strictly decreasing function of  $G_\Lambda(X - Y)$  (the common factor  $S_\Lambda > 0$  is immaterial to order), so its ranks converge to those of  $-G(X - Y)$ ; atomlessness of the limit law makes the Spearman and Kendall functionals continuous there, giving convergence. On  $\mathbb{S}^1$ ,  $G(\theta) = \sum_{k \geq 1} \cos(k\theta)/k^2$  is strictly decreasing on  $[0, \pi]$  (its derivative  $-\sum_{k \geq 1} \sin(k\theta)/k = (\theta - \pi)/2 < 0$  there), so  $-G$  is strictly increasing in the geodesic distance  $|\theta|$  and the rank correlation is 1.  $\square$

This upgrades the empirical “flat in  $m$ ” (§5.2) to a rank-stability theorem: for  $d \leq 3$  the commute angular ranking converges to a fixed Green-kernel ranking—exact on  $\mathbb{S}^1$ , and a computable limiting quantity on  $\mathbb{T}^2, \mathbb{T}^3$  (where  $G$  is direction-dependent, so the limit is below 1). For  $d \geq 4$  the kernel is not  $L^2$ -stable (Proposition SM2.5) and no such limit is claimed.

**The general Green-kernel rank limit** ( $d \in \{2, 3\}$ ). The flat-torus computation generalizes to every closed manifold of dimension 2 or 3, bypassing the impossible uniform upper Lipschitz bound and isolating the exact geometric obstruction. For independent  $X, Y$  from the volume measure, define the *Green-rank coefficient*  $\rho_G(M) := \rho_S(d_M(X, Y), -G_M(X, Y))$ , where  $G_M$  is the zero-mean Green kernel of the Laplace–Beltrami operator. Angular Preservation then splits into two questions: does the truncated angular ranking converge uniformly in  $m$  to the Green-kernel ranking, and is  $\rho_G(M) > 0$ ?

**THEOREM SM2.7** (Green-kernel limit of the commute angular ranking). *Let  $M$  be a closed connected Riemannian manifold of dimension  $d \in \{2, 3\}$ , volume normalized to 1, with Laplace–Beltrami eigenpairs  $(\mu_k, \varphi_k)$ . At a multiplet-closed cutoff  $\Lambda$  set  $G_\Lambda(x, y) = \sum_{0 < \mu_k \leq \Lambda} \varphi_k(x) \varphi_k(y) / \mu_k$ ,  $S_\Lambda(x) = G_\Lambda(x, x)$ , and  $N_\Lambda(x) = (\mu_k^{-1/2} \varphi_k(x))_{0 < \mu_k \leq \Lambda} / \sqrt{S_\Lambda(x)}$ . Let  $(X, Y)$  have bounded density with respect to  $\text{vol} \otimes \text{vol}$ , with the laws of  $d_M(X, Y)$  and  $G(X, Y)$  atomless, where  $G(x, y) = \sum_{k \geq 1} \varphi_k(x) \varphi_k(y) / \mu_k \in L^2(M \times M)$ . Then*

$$\rho_S(d_M(X, Y), \|N_\Lambda(X) - N_\Lambda(Y)\|) \rightarrow \rho_S(d_M(X, Y), -G(X, Y)),$$

and likewise for Kendall’s  $\tau$ . Consequently, if  $\rho_G(M) > 0$  there are  $\Lambda_0 < \infty$  and  $\rho_0 > 0$  with  $\rho_S(d_M(X, Y), \|N_\Lambda(X) - N_\Lambda(Y)\|) \geq \rho_0$  for every multiplet-closed  $\Lambda \geq \Lambda_0$ : the rank form of Angular Preservation is uniform in the mode count beyond a finite cutoff.

*Proof.* Write  $C_\Lambda(x, y) = \langle N_\Lambda(x), N_\Lambda(y) \rangle = G_\Lambda(x, y) / \sqrt{S_\Lambda(x) S_\Lambda(y)}$ ; since  $\|N_\Lambda(x) - N_\Lambda(y)\|^2 = 2 - 2C_\Lambda(x, y)$ , angular distance carries the reverse ranking of  $C_\Lambda$ . (1) *Hilbert–Schmidt limit.* The  $\varphi_k \otimes \varphi_k$  are orthonormal in  $L^2(M \times M)$ , so  $\|G_{\Lambda'} - G_\Lambda\|_{L^2}^2 = \sum_{\Lambda < \mu_k \leq \Lambda'} \mu_k^{-2}$ ; by Weyl’s law  $\mu_k \asymp k^{2/d}$  this converges iff  $d < 4$ , so  $G_\Lambda \rightarrow G$  in  $L^2(M \times M)$  for  $d = 2, 3$ . (2) *The diagonal is asymptotically constant.* The uniform pointwise Weyl law  $\sum_{\mu_k \leq L} \varphi_k(x)^2 = c_d L^{d/2} + O(L^{(d-1)/2})$  and partial summation give, uniformly in  $x$ ,  $S_\Lambda(x) = a_2 \log \Lambda + O(1)$  ( $d = 2$ ) or  $a_3 \Lambda^{1/2} + O(\log \Lambda)$  ( $d = 3$ ); writing  $A_\Lambda$  for the leading term,  $\sup_x |S_\Lambda(x) / A_\Lambda - 1| \rightarrow 0$ . (3) *Rescale.*  $H_\Lambda := A_\Lambda C_\Lambda = G_\Lambda \cdot A_\Lambda / \sqrt{S_\Lambda(x) S_\Lambda(y)} \rightarrow G$  in  $L^2(M \times M)$  (the second factor  $\rightarrow 1$  uniformly, and  $\sup_\Lambda \|G_\Lambda\|_2 < \infty$ ). Multiplying  $C_\Lambda$  by the positive constant  $A_\Lambda$  leaves ranks unchanged, so  $\text{rank}\|N_\Lambda(X) - N_\Lambda(Y)\| = \text{rank}(-H_\Lambda(X, Y))$ . (4) *Rank convergence.* A bounded pair density carries  $H_\Lambda \rightarrow G$  in  $L^2$  to convergence in probability under the law of  $(X, Y)$ ; atomlessness of the limit makes the distribution functions converge, and the population Spearman/Kendall identities give  $\rho_S(d_M, -H_\Lambda) \rightarrow \rho_S(d_M, -G)$ .  $\square$

**COROLLARY SM2.8** (Uniform rank preservation on two-point homogeneous spaces). *If, in addition,  $M$  is compact two-point homogeneous, then  $G(x, y) = g(d_M(x, y))$  with  $g$  strictly decreasing on  $(0, \text{diam } M)$ ; hence  $\rho_G(M) = 1$  and  $\rho_S(d_M(X, Y), \|N_\Lambda(X) - N_\Lambda(Y)\|) \rightarrow 1$ , bounded below uniformly beyond a finite cutoff. This proves the eventual uniform-in- $m$  rank form of Conjecture 3.4 on  $S^2, S^3, \mathbb{RP}^2, \mathbb{RP}^3$  (the circle being the exact Fourier case above).*

*Proof.* Two-point homogeneity makes the invariant kernel a function of geodesic distance alone. With  $A(r)$  the area density of a geodesic sphere, the zero-mean Green function satisfies  $\Delta g = -1/\text{vol } M$  off the pole, i.e.  $A(r)g'(r) = -1 + \text{vol}(M)^{-1} \int_0^r A(s) ds$ ; since  $\int_0^r A < \text{vol } M$  for  $r < \text{diam } M$ ,  $g'(r) < 0$ , so  $-G$  is strictly increasing in  $d_M$  and the population rank correlation is 1.  $\square$

*Graph transfer of the shrinking signal.* The rank-carrying angular fluctuation has amplitude  $\asymp A_\Lambda^{-1/2}$ , so the growing-window transfer condition sharpens. With the calibration error  $E_{n,m} = \max_i \|Q_n \tilde{\Psi}_i - F_m(x_i)\|$  of Appendix SM1 and  $R_m \asymp \sqrt{A_m}$ , normalization stability gives  $|A_m \langle \hat{q}_i, \hat{q}_j \rangle - A_m \langle N_m(x_i), N_m(x_j) \rangle| \leq C E_{n,m} \sqrt{A_m}$ . The rank-transfer condition is therefore  $E_{n,m} \sqrt{A_m} \rightarrow 0$ —that is,  $E_{n,m} \sqrt{\log \mu_m} \rightarrow 0$  ( $d = 2$ ) or  $E_{n,m} \mu_m^{1/4} \rightarrow 0$  ( $d = 3$ ): graph calibration must converge *faster* than the rank-carrying signal shrinks, sharpening the bare  $E_{n,m} \rightarrow 0$ .

CONJECTURE SM2.9 (Green-rank positivity). *For the stated geometric class and pair law,  $\rho_G(M) = \rho_S(d_M(X, Y), -G_M(X, Y)) > 0$ .*

This reorganizes the frontier. The *rank* component is proved: for  $d \leq 3$  the commute angular ranking converges uniformly in the cutoff to the Green-kernel ranking (Theorem SM2.7), positive and equal to 1 on the circle, flat tori (with a computable coefficient), and two-point homogeneous spaces of dimension  $\leq 3$  (Corollary SM2.8). What remains is the purely geometric question of Green-rank positivity, and the exact boundary is  $d = 4$ , where  $\sum_k \mu_k^{-2} = \infty$  and the commute Green kernel ceases to be Hilbert-Schmidt (a stronger filter  $\mu_k^{-s/2}$ ,  $s > d/4$ , restores the limit). In one line: uniform-in-mode *metric* control fails for the commute embedding, but uniform-in-mode *rank* control has a Green-kernel limit below the critical dimension  $d = 4$ , and Angular Preservation reduces to positivity of a geometric coefficient—settled on low-dimensional two-point homogeneous spaces.

**Proof of Proposition 3.17 (concentration).** *Plain filter.* With  $E_\Lambda(x, y) = \sum_{0 < \mu_k \leq \Lambda} \varphi_k(x) \varphi_k(y)$  (constant mode removed),

$$\|F_\Lambda^{(1)}(x) - F_\Lambda^{(1)}(y)\|^2 = E_\Lambda(x, x) + E_\Lambda(y, y) - 2E_\Lambda(x, y), \quad R_\Lambda^{(1)}(x)^2 = E_\Lambda(x, x).$$

The pointwise Weyl law with uniform remainder [SM9] gives  $E_\Lambda(x, x) = C_d \Lambda^{d/2} + O(\Lambda^{(d-1)/2})$  with spatially constant leading coefficient, and for  $d_M(x, y) \geq \delta$  the off-diagonal spectral function obeys  $|E_\Lambda(x, y)| = O(\Lambda^{(d-1)/2})$  uniformly, oscillating in the pair separation on the wavelength scale  $\Lambda^{-1/2}$  (the Bessel-type scaling limit of the spectral projector [SM7]). The radial difference is  $R_\Lambda^{(1)}(x) - R_\Lambda^{(1)}(y) = (E_\Lambda(x, x) - E_\Lambda(y, y)) / (R_\Lambda^{(1)}(x) + R_\Lambda^{(1)}(y)) = O(\Lambda^{(d-1)/2-d/4})$ , so its square is  $O(\Lambda^{d/2-1})$ . Feeding these into the identity of Lemma 3.6,

$$\|N_\Lambda^{(1)}(x) - N_\Lambda^{(1)}(y)\|^2 = \frac{2C_d \Lambda^{d/2} + O(\Lambda^{(d-1)/2})}{C_d \Lambda^{d/2} (1 + O(\Lambda^{-1/2}))} = 2 + O(\Lambda^{-1/2})$$

uniformly over separated pairs, with pair-dependence carried by  $-2E_\Lambda(x, y)/E_\Lambda(x, x)$ . The numerator kernel oscillates with sign changes on the wavelength scale, so it induces no monotone ordering in  $d_M$ . Concentration by itself, however, does not force the rank correlation to zero (a shrinking correction can preserve order); the exact collapse of both the Spearman and Kendall correlations against geodesic distance is established on the circle—where  $E_\Lambda$  is the Dirichlet kernel—in Theorem 3.18 below.

This is the continuum mechanism behind the plain-eigenmap rank separation in the weighting ablation (§5.3).

*Commutate filter.* The same computation with weights  $\mu_k^{-1}$  replaces  $E_\Lambda$  by  $G_\Lambda(x, y) = \sum_{0 < \mu_k \leq \Lambda} \mu_k^{-1} \varphi_k(x) \varphi_k(y)$ : the diagonal is  $C_d' \Lambda^{d/2-1}$  ( $d > 2$ ;  $C_2 \log \Lambda$ ,  $d = 2$ ) with spatially constant leading coefficient (Step 7 of the Proposition 3.16 proof), the radial term is again lower order, and

$$\|N_\Lambda(x) - N_\Lambda(y)\|^2 = 2 - \frac{2G_\Lambda(x, y)}{G_\Lambda(x, x)}(1 + o(1)).$$

The correction kernel decays like  $\mu^{-1}$ , and for  $d \leq 3$  (where  $\sum_k \mu_k^{-2} < \infty$ ) it converges off the diagonal, in  $L^2$ , to the Green's function of  $M$ —a distance-structured kernel, and on a *two-point homogeneous* (isotropic) space a function of  $d_M$  alone. For  $d \geq 4$  the sharp truncated Green kernel is no longer Hilbert–Schmidt stable and high-frequency oscillations can enter, so we claim no low-frequency dominance there. Both filters therefore concentrate at fixed scale as  $\Lambda \rightarrow \infty$  (the wavelength story of Proposition 3.16), but the *ordering* information that rank statistics read is carried by a coherent kernel for commute and an oscillatory one for plain: the ablation's rank separation, in the continuum.

**Proof of Theorem 3.18 (exact plain-filter rank collapse on  $\mathbb{S}^1$ ).** Write  $\theta = x - y$ ; by evenness the chord depends only on the geodesic distance  $|\theta| \in [0, \pi]$ . Since  $\|F_N\|^2 \equiv N$ ,

$$\langle \widehat{F}_N(x), \widehat{F}_N(y) \rangle = \frac{1}{N} \sum_{k=1}^N \cos(k\theta), \quad Q_N(\theta) = 2 - \frac{2}{N} Z_N(\theta), \quad Z_N(\theta) := \sum_{k=1}^N \cos(k\theta).$$

The Dirichlet-kernel identity  $1 + 2 \sum_{k=1}^N \cos(k\theta) = \sin((N + \frac{1}{2})\theta) / \sin(\theta/2)$  gives, with  $a(\theta) = 1/(2 \sin(\theta/2))$  and  $\omega_N = N + \frac{1}{2}$ ,

$$W_N(\theta) := Z_N(\theta) + \frac{1}{2} = a(\theta) \sin(\omega_N \theta).$$

As  $Q_N$  is a strictly decreasing affine function of  $Z_N$ , and  $Z_N, W_N$  differ by a constant, the three share the same rank correlations with  $\Theta$  up to a global sign; it suffices to prove  $\rho_S(\Theta, W_N(\Theta)) \rightarrow 0$  and  $\tau_K(\Theta, W_N(\Theta)) \rightarrow 0$ .

*Step 1 (two-scale equidistribution).* Fix  $\varepsilon > 0$ ; on  $[\varepsilon, \pi]$  the envelope  $a$  is continuous and bounded. For any  $\varphi(\theta, t)$  bounded, continuous, and  $2\pi$ -periodic in  $t$ ,

$$\frac{1}{\pi} \int_\varepsilon^\pi \varphi(\theta, \omega_N \theta) d\theta \rightarrow \frac{1}{\pi} \int_\varepsilon^\pi \frac{1}{2\pi} \int_0^{2\pi} \varphi(\theta, t) dt d\theta,$$

by expanding  $\varphi$  in Fourier modes in  $t$ : the zeroth mode is the phase average, and each nonzero mode is an oscillatory integral  $\int_\varepsilon^\pi c_m(\theta) e^{im\omega_N \theta} d\theta \rightarrow 0$  by Riemann–Lebesgue as  $\omega_N \rightarrow \infty$ . Hence  $(\Theta, W_N(\Theta)) \Rightarrow (\Theta, a(\Theta) \sin T)$  with  $T \sim \text{Unif}[0, 2\pi]$  independent of  $\Theta$  (the removed collar  $[0, \varepsilon)$  has probability  $\varepsilon/\pi \rightarrow 0$ , and  $a(\Theta) < \infty$  a.s.). The limit  $W := a(\Theta) \sin T$  is atomless and symmetric, so its CDF  $H$  is continuous with  $H(-z) = 1 - H(z)$ .

*Step 2 (Spearman).* For continuous variables the population Spearman coefficient is  $\rho_S(U, V) = 12 \mathbb{E}[F_U(U)F_V(V)] - 3$ . With  $F_\Theta(\Theta) = \Theta/\pi$  and  $H_N$  the CDF of  $W_N(\Theta)$ ,

$$\rho_S(\Theta, W_N(\Theta)) = 12 \mathbb{E}\left[\frac{\Theta}{\pi} H_N(W_N(\Theta))\right] - 3.$$

Weak convergence to the atomless  $H$  gives  $\sup_z |H_N(z) - H(z)| \rightarrow 0$  (Pólya), so  $H_N$  may be replaced by  $H$ . Step 1 then yields

$$\mathbb{E}\left[\frac{\Theta}{\pi} H(W_N(\Theta))\right] \longrightarrow \frac{1}{\pi} \int_0^\pi \frac{\theta}{\pi} \left( \frac{1}{2\pi} \int_0^{2\pi} H(a(\theta) \sin t) dt \right) d\theta.$$

Pairing  $t$  with  $t + \pi$  gives  $H(a \sin t) + H(-a \sin t) = 1$ , so the inner average is  $\frac{1}{2}$  and the outer integral is  $\frac{1}{4}$ . Hence  $\rho_S(\Theta, W_N(\Theta)) \rightarrow 12 \cdot \frac{1}{4} - 3 = 0$ , and by the affine reversal  $\rho_S(\Theta, Q_N(\Theta)) \rightarrow 0$ .

*Step 3 (Kendall).* Take independent copies  $(\Theta_1, \Theta_2)$  and write  $\tau_K = \mathbb{E}[\text{sgn}(\Theta_1 - \Theta_2) \text{sgn}(W_N(\Theta_1) - W_N(\Theta_2))]$ . Applying the Fourier/Riemann–Lebesgue argument of Step 1 jointly, the four-tuple  $(\Theta_1, \Theta_2, \omega_N \Theta_1, \omega_N \Theta_2)$  converges weakly to  $(\Theta_1, \Theta_2, T_1, T_2)$  with  $T_1, T_2 \sim \text{Unif}[0, 2\pi]$  independent of one another and of  $(\Theta_1, \Theta_2)$  (each nonzero joint Fourier mode is killed by integrating first in whichever variable carries nonzero frequency). The integrand  $\text{sgn}(\theta_1 - \theta_2) \text{sgn}(a(\theta_1) \sin t_1 - a(\theta_2) \sin t_2)$  is bounded and its discontinuity set  $\{\theta_1 = \theta_2\} \cup \{a(\theta_1) \sin t_1 = a(\theta_2) \sin t_2\}$  is null under the limit law, so the portmanteau theorem gives  $\tau_K \rightarrow \mathbb{E}[\text{sgn}(\Theta_1 - \Theta_2) \text{sgn}(a(\Theta_1) \sin T_1 - a(\Theta_2) \sin T_2)]$ . Conditioned on  $(\Theta_1, \Theta_2)$  the inner sign is that of a difference of two independent atomless symmetric variables, so its expectation is 0; hence  $\tau_K(\Theta, W_N(\Theta)) \rightarrow 0$  and  $\tau_K(\Theta, Q_N(\Theta)) \rightarrow 0$ .

**Graph transfer at growing  $m$ , and the  $C^1$  route.** For fixed  $m$ , Appendix SM1 is the complete transfer. For a growing spectral window  $m = m(n)$ , the same calibration goes through—with every constant carrying an  $m$ -subscript—whenever

$$\mathcal{E}_n(\mu_m) \gamma_m^{-1} \longrightarrow 0,$$

where  $\gamma_m$  is the spectral gap isolating the retained block and  $\mathcal{E}_n(\mu_m)$  is the graph-Laplacian consistency error on continuum eigenfunctions up to frequency  $\mu_m$ ; since high-frequency eigenfunctions have polynomially large derivatives,  $\mathcal{E}_n(\mu_m)$  grows polynomially in  $\mu_m$ , so  $m(n)$  may grow, but not too fast. The resulting cutoff is  $\theta_{n,m} \gtrsim E_{n,m}/(a_m \rho_m^-)$ , every factor now  $m$ -dependent. Separately, the additive pairwise error of the sup-norm argument is what forces  $\theta_n$  to sit at the spectral-rate scale; a  $C^{0,1}$  eigenvector theory—the Lipschitz estimates of [SM6] are the available ingredient—would replace it by a multiplicative derivative-level error ( $\|d\hat{Z}(v)\| \geq (1 - \eta)\|dN(v)\|$  once the  $C^1$  calibration error is below  $\eta a_m \rho_m^-$ ), pushing the cutoff down to the connectivity scale  $\varepsilon_n$ —optimal, since below it combinatorial twins exist (Appendix SM1). We state these as the conditions a growing-window theorem needs, not as a theorem.

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